

Leverage, Endogenous Unbalanced Growth, and Asset Price Bubbles¹

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¹Link to paper: <https://arxiv.org/abs/2211.13100> 

Asset price bubbles

- Bubble means asset price exceeds fundamental value:

$$P_t > V_t, \quad V_t = \text{present value of dividends/rents } D_t.$$

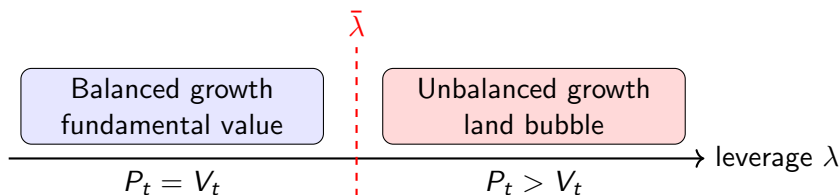
- Large literature on “pure bubble” ($D = 0$), but many shortcomings:
 - lack of realism,
 - equilibrium indeterminacy (continuum of equilibria),
 - inability to connect to empirical “bubble detection” literature
- Theory of rational bubble with $D > 0$ only recent
 - Hirano and Toda (2025a, JPE): bubble necessity under $R < G_d < G$
 - Hirano and Toda (2025c, PNAS): unbalanced technological growth and land bubble
 - Review: Hirano and Toda (2024, JME)

Financial innovation and asset boom

- Some document connection between financial innovation and asset boom (Fostel and Geanakoplos, 2012; Scheinkman, 2014; Jordà et al., 2015; Mian and Sufi, 2015; Quinn and Turner, 2020)
- Kindleberger and Aliber (2005, Ch. 4):
Axiom number two. Asset price bubbles depend on the growth of credit. [...] every mania has been associated with the expansion of credit.
- Rational (pure) bubble literature since 2008 has modeled role of financial conditions, but typically implies bubbles more likely when financial conditions **tighter**, not looser

This paper

- Build macro-finance model of **bubbles attached to land** (dividend-paying asset)
- Key parameter: financial leverage λ
- Main result: leverage creates **phase transition**:



Message

Loose financial conditions can make rational bubbles in real assets **necessary**, not merely possible

Related literature

- **Financial accelerator**: Greenwald and Stiglitz (1993), Kiyotaki and Moore (1997), Lorenzoni (2008), Brunnermeier and Sannikov (2014) etc.
- **Unbalanced growth**: Baumol (1967), Matsuyama (1992), Buera and Kaboski (2012), etc.
- **Rational bubbles**: Samuelson (1958), Bewley (1980), Tirole (1985), Scheinkman and Weiss (1986), Kocherlakota (1992), Hirano and Toda (2024, 2025b), etc.
- **Necessity of bubbles**: Wilson (1981), Hirano and Toda (2025a,c)

Model overview

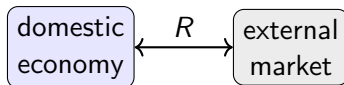
- Key features of model:
 - Infinite horizon: $t = 0, 1, \dots$
 - Infinitely-lived heterogeneous agents
 - Incomplete markets
 - Leverage constraint
 - AK model at individual level, neoclassical production with land at aggregate level
- For exposition and transparency, focus on “large open” economy

Why start with large open economy?

- “Open”: agents can lend to and borrow from external source at constant gross rate R
- “Large”: aggregate external savings or debt asymptotically negligible:

$$\frac{B_t}{W_t} \rightarrow 0 \quad \text{as } t \rightarrow \infty$$

- This trick keeps model analytically tractable while retaining general equilibrium discipline



Agents and idiosyncratic productivity

- Continuum of infinitely-lived agents indexed by $i \in [0, 1]$
- Log utility:

$$E_0 \sum_{t=0}^{\infty} \beta^t \log c_t, \quad 0 < \beta < 1$$

- Each period, agents draw idiosyncratic investment productivity

$$z_t \sim \Phi, \quad \text{independently across agents and time}$$

- Log utility implies simple consumption rule

$$c_t = (1 - \beta)w_t$$

Technology and land

Linear capital technology

- If agent with productivity z_t invests i_t units, $k_{t+1} = z_t i_t$
- One unit of capital yields $m > 0$ units of good

Land

- Land in unit supply and pays constant rent $D > 0$
- No-arbitrage pricing:

$$P_t = \frac{1}{R}(P_{t+1} + D)$$

- Fundamental value when $R > 1$:

$$V = \frac{D}{R - 1}$$

Land is a real, dividend-paying asset, not pure bubble

Budget and leverage

- At time t , agent with wealth w_t chooses consumption, investment, and asset holdings:

$$c_t + i_t + a_t = w_t$$

- Next-period wealth is

$$w_{t+1} = \underbrace{mz_t i_t}_{\text{capital income}} + \underbrace{Ra_t}_{\text{asset income}}$$

- Leverage constraint:

$$0 \leq i_t \leq \lambda(i_t + a_t), \quad \lambda \geq 1$$

- Equivalently,

$$\underbrace{-a_t}_{\text{borrowing}} \leq \underbrace{(1 - 1/\lambda) i_t}_{\text{collateral ratio}}$$

Individual portfolio choice

- Return on investment: mz
- Return on land/net saving: R
- Investment cutoff:

$$mz > R \iff z > \bar{z} := R/m$$

Optimal allocation after saving βw_t

$$(i_t, a_t) = \begin{cases} (0, \beta w_t) & \text{if } z_t \leq \bar{z}, \\ (\lambda \beta w_t, (1 - \lambda) \beta w_t) & \text{if } z_t > \bar{z} \end{cases}$$

Productive agents borrow and invest; unproductive agents save.

Aggregate wealth growth

- Individual wealth evolves as

$$w_{t+1} = \beta (R + \lambda \max\{0, mz_t - R\}) w_t$$

- Aggregating over continuum of agents gives

$$W_{t+1} = \beta (R + \lambda \pi(R)) W_t,$$

where

$$\pi(R) := \int_0^\infty \max\{0, mz - R\} d\Phi(z)$$

is expected excess return on unlevered capital investment

Growth factor

$$G = \beta (R + \lambda \pi(R))$$

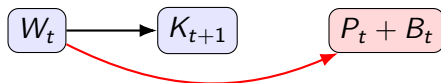
Aggregating capital and land demand

- Aggregate capital:

$$K_{t+1} = \beta \lambda W_t \int_{\bar{z}}^{\infty} z \, d\Phi(z).$$

- Aggregate land plus external saving:

$$\begin{aligned} P_t + B_t &= \beta W_t \int_0^{\bar{z}} d\Phi(z) + \beta(1 - \lambda) W_t \int_{\bar{z}}^{\infty} d\Phi(z) \\ &= \beta(\lambda \Phi(\bar{z}) + 1 - \lambda) W_t. \end{aligned}$$



Trend stationary equilibrium

A trend stationary equilibrium consists of

$$R, \quad G, \quad \bar{z}, \quad \{W_t, K_{t+1}, P_t, B_t\}_{t \geq 0}$$

satisfying:

- no-arbitrage on land, $P_t = (P_{t+1} + D)/R$;
- growth condition, $G = \beta(R + \lambda\pi(R))$;
- cutoff, $\bar{z} = R/m$;
- aggregate capital and land-market equations;
- large-economy condition, $B_t/W_t \rightarrow 0$.

Fundamental value in this model

$$V_t = \sum_{s=1}^{\infty} R^{-s} D = \frac{D}{R-1}$$

Fundamental versus bubbly pricing

Fundamental

$$P_t = V_t = \frac{D}{R-1}$$

Land price is pinned down by rents

- In model, rents are constant: $G_d = 1$
- If economy grows at $G > 1$, land demand grows but rents do not

Bubbly

$$P_t > V_t$$

Land price contains a speculative component

Critical leverage threshold

- Threshold for phase transition comes from knife-edge case $G = R = 1$:

$$G = \beta(R + \lambda\pi(R)) \implies 1 = \beta(1 + \lambda\pi(1)).$$

- Solving for leverage yields

$$\bar{\lambda} = \frac{1 - \beta}{\beta} \frac{1}{\int_{1/m}^{\infty} (mz - 1) d\Phi(z)}$$

- A thicker productive tail or higher m lowers leverage needed for takeoff

Land bubble characterization and phase transition

Theorem

Let $\bar{\lambda}$ be leverage threshold. Then

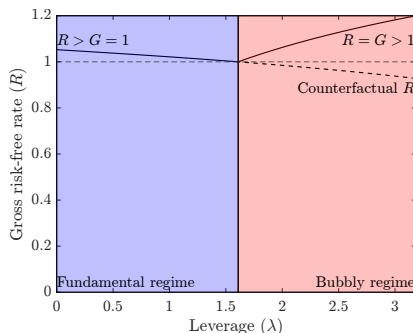
1. If $\lambda < \bar{\lambda}$, we have $P_t = V_t$ for all t . The economy exhibits balanced growth and the price-rent ratio is constant.
 2. If $\lambda > \bar{\lambda}$, we have $P_t > V_t$ for all t . The economy exhibits unbalanced growth and the price-rent ratio diverges to ∞ .
- Economy exhibits **phase transition**
 - Low λ : exogenous growth ($G = 1$), fundamental value
 - High λ : endogenous growth ($G > 1$), asset bubble

V-shaped interest rate

Proposition

Equilibrium gross risk-free rate R is strictly decreasing (increasing) for $\lambda < \bar{\lambda}$ ($\lambda > \bar{\lambda}$).

- In fundamental regime, as leverage relaxed, marginal investor becomes less productive, hence low interest rate
- In bubbly regime, as leverage relaxed, $G = R \uparrow$



Bubble necessity

Hirano and Toda (2025a) give **bubble necessity** condition:

$$R < G_d < G,$$

where

- R : counterfactual (natural) interest rate without asset
- G_d : dividend growth rate
- G : economic growth rate

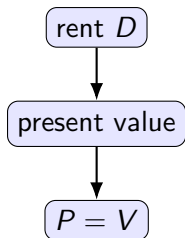
In large open economy, above threshold $\bar{\lambda}$,

$$R < G_d = 1 < G$$

Finite fundamental value cannot support required demand for land;
equilibrium price must contain a bubble

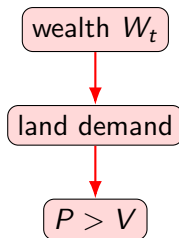
Supply-driven vs demand-driven land prices

Fundamental regime



Supply factor disciplines price

Bubbly regime

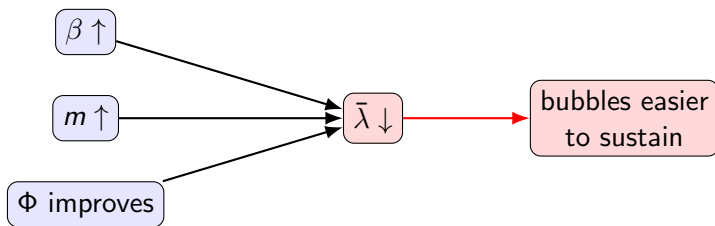


Demand growth sustains price

Comparative statics of leverage threshold

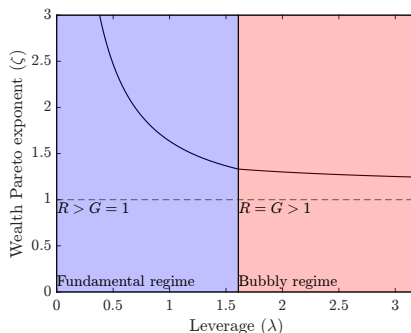
$$\bar{\lambda}(\beta, m, \Phi) = \frac{1 - \beta}{\beta} \frac{1}{E[(mZ - 1)_+]}$$

- Greater patience β lowers $\bar{\lambda}$.
- Higher marginal product m lowers $\bar{\lambda}$.
- Productivity distribution Φ that improves by first-order stochastic dominance lowers $\bar{\lambda}$.



Bubbles and wealth inequality

- Add birth/death (perpetual youth); use Beare and Toda (2022, ECMA) formula to compute Pareto exponent
- In fundamental regime, as leverage relaxed, unproductive agents earn low $R \rightarrow$ higher wealth inequality
- In bubbly regime, as leverage relaxed, $R \uparrow$ and unproductive catch up



From simple to general model

Production

Let f be neoclassical production function (e.g., CES),

$$F(K, X) = f(K, X) + (1 - \delta)K,$$
$$\lim_{K \rightarrow \infty} \frac{F(K, 1)}{K} = \lim_{K \rightarrow \infty} F_K(K, 1) = m > 0$$

Land and REIT

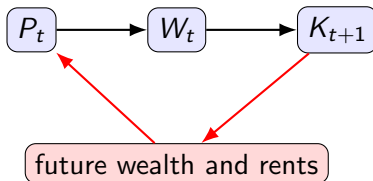
Land rent is endogenous and land is risk-free because there is no aggregate uncertainty:

$$r_t = F_X(K_t, 1), \quad R_t = \frac{P_{t+1} + F_X(K_{t+1}, 1)}{P_t}$$

Financial accelerator

Aggregate wealth is

$$W_t = F_K(K_t, 1)K_t + P_t + F_X(K_t, 1) = F(K_t, 1) + P_t$$



- Higher land prices raise current wealth
- Higher wealth raises next-period capital
- Higher capital and wealth feed back into future land prices

Asymptotic behavior

Equilibrium can be reduced to two-dimensional dynamics in $(K_t, \bar{z}_t)$

$$\lambda < \bar{\lambda}$$

- Dynamics has steady state
- Convergence:

$$K_t \rightarrow K^*, \quad P_t \rightarrow P^*$$

- Balanced growth and no land bubble

$$\lambda > \bar{\lambda}$$

- No steady state
- But a unique $G > 1$ solves

$$\frac{G}{m} = \frac{\beta \lambda \int_{G/m}^{\infty} z \, d\Phi(z)}{1 - \beta(\lambda \Phi(G/m) + 1 - \lambda)}$$

Why growth is unbalanced

- Suppose economy grows:

$$K_t \sim kG^t, \quad W_t \sim wG^t, \quad P_t \sim pG^t, \quad G > 1$$

- Land rent is

$$r_t = F_X(K_t, 1)$$

- If $\sigma > 1$ denotes K - X elasticity, then $r_t \sim rG^{t/\sigma}$

Implication

$$\frac{P_t}{r_t} \rightarrow \infty$$

Land price grows with aggregate wealth; pure land rent grows more slowly $\implies$ land bubble

Which elasticity matters?

The relevant elasticity is the elasticity of substitution in aggregate wealth F :

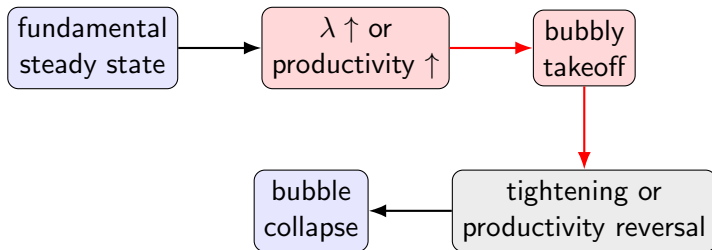
$$\sigma(K, X) = -\frac{\partial \log(K/X)}{\partial \log(F_K/F_X)}$$

- If F has positive asymptotic slope m , then asymptotically $\sigma \geq 1$.
- For CES production $f(K, X) = A(\alpha K^{1-\rho} + (1-\alpha)X^{1-\rho})^{\frac{1}{1-\rho}}$,

$$\lim_{K \rightarrow \infty} \sigma(K, 1) = \begin{cases} 1/\rho & \text{if } \rho < 1, \\ 1/\alpha & \text{if } \rho = 1, \\ \infty & \text{if } \rho > 1 \end{cases}$$

- With $\sigma > 1$ and high leverage, growth equilibrium is unique for sufficiently large K_0

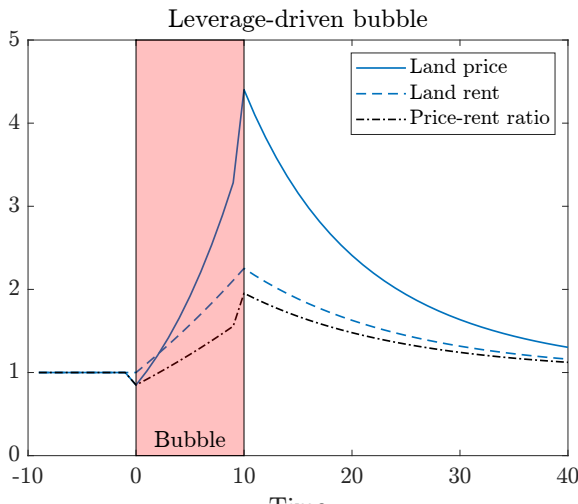
Boom–bust interpretation



- During bubbly episode, capital, wealth, consumption, and land prices rise
- Land prices grow faster than rents, so price-rent ratio rises
- If leverage or productivity reverses, bubble collapses and price-rent ratio falls

Leverage-driven bubble

- Set $\rho = 1$ (Cobb-Douglas)
- Change leverage λ from low to high, then back to low (unanticipated)



Japan's bubble economy

Takeoff forces

- financial deregulation in early 1980s;
- lower discount rate after yen appreciation (Plaza accord);
- optimism about Tokyo as financial hub;
- capital flowing into real estate

Collapse forces

- concern over the “land problem”;
- Bank of Japan rate hikes;
- quotas on real-estate lending;
- land value tax

Model interprets this as unexpected transition into and out of bubbly regime

Concluding remarks

- Proposed macro-finance theory to think about bubbles attached to dividend-paying assets
 - Model circumvents shortcomings of pure bubble models including lack of realism, equilibrium indeterminacy, and inability to connect to empirical literature
- Balanced or unbalanced growth endogenously determined by financial leverage:
 - Low leverage: weak financial accelerator, balanced growth, fundamental value
 - High leverage: strong financial accelerator, unbalanced growth, bubble
- Hence with financial (or technological) development and presence of fixed factor like land, bubbles become inevitable

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Figure: Japanese bubble in late 1980s

皇居の敷地



アメリカ
カリフォルニア州
全部の不動産

HOLLY
WOOD

>



or

カナダの
不動産全部



日本の
不動産全部



アメリカの
不動産全部



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※アメリカの面積は
日本の面積の25倍。